

Name \_\_\_\_\_ Period \_\_\_\_\_

**Skill 39.1 Exercise 1**

For each scenario, decide whether it sounds more like *traditional programming* or *machine learning*.

| Scenario                                                                                                      | Traditional Programming<br>or Machine Learning? | Explain |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------|
| A program calculates the average of five test scores using a formula.                                         |                                                 |         |
| A program learns from thousands of emails marked spam or not spam, then predicts whether a new email is spam. |                                                 |         |
| A program prints “eligible” if a person is at least 18 years old.                                             |                                                 |         |
| A music app studies listening history and recommends songs a user might like.                                 |                                                 |         |

**Skill 39.2 Exercise 1**

A streaming app has the following dataset.

| minutes_watched | number_of_likes | previous_genre | liked_new_show |
|-----------------|-----------------|----------------|----------------|
| 120             | 8               | comedy         | yes            |
| 45              | 1               | drama          | no             |
| 200             | 12              | comedy         | yes            |
| 80              | 3               | action         | no             |

What is the label?

What are three possible features?

What does each row represent?

Why would the app want to train a model using this data?

Name \_\_\_\_\_ Period \_\_\_\_\_

**Skill 39.3 Exercise 1**

For each dataset, decide whether it is **labeled** or **unlabeled**.

| Dataset Description                                                                              | Labeled or Unlabeled? | Explain |
|--------------------------------------------------------------------------------------------------|-----------------------|---------|
| Emails with sender, number of links, subject length, and spam/not spam.                          |                       |         |
| Customer shopping data with purchase totals and visit frequency, but no customer group assigned. |                       |         |
| Houses with square footage, number of bedrooms, and sale price.                                  |                       |         |
| Songs with tempo, energy, danceability, and no genre label.                                      |                       |         |

**Skill 39.4 Exercise 1**

For each situation, identify whether it is **supervised learning** or **unsupervised learning**.

| Scenario                                                                                                 | Supervised or Unsupervised? | Explain |
|----------------------------------------------------------------------------------------------------------|-----------------------------|---------|
| A model predicts whether a patient has a disease using past patient records that include disease status. |                             |         |
| A company groups customers into unknown categories based on shopping behavior.                           |                             |         |
| A model predicts the sale price of a used car using data from cars that already sold.                    |                             |         |
| A social media app finds groups of users with similar viewing habits.                                    |                             |         |

**Skill 39.5 Exercise 1**

For each task, identify whether it is **regression**, **classification**, or **clustering**.

Name \_\_\_\_\_ Period \_\_\_\_\_

| Task                                                                              | Regression, Classification, or Clustering? | Explain |
|-----------------------------------------------------------------------------------|--------------------------------------------|---------|
| Predicting the number of points a basketball player will score.                   |                                            |         |
| Predicting whether a customer will cancel a subscription: yes or no.              |                                            |         |
| Grouping students based on study habits without knowing the groups ahead of time. |                                            |         |
| Predicting the price of a used phone.                                             |                                            |         |
| Predicting whether a review is positive, negative, or neutral.                    |                                            |         |

#### Skill 39.6 Exercise 1

For each scenario, identify:

- whether the data is labeled or unlabeled
- whether the problem is supervised or unsupervised
- whether the task is regression, classification, or clustering

| Scenario                                                                                                     | Labeled or Unlabeled? | Supervised or Unsupervised? | ML Task |
|--------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------|---------|
| A real estate website predicts house prices using past home sales.                                           |                       |                             |         |
| A school predicts whether students are likely to pass a course using past student records labeled pass/fail. |                       |                             |         |
| A company groups customers based on shopping habits without any pre-existing group names.                    |                       |                             |         |
| A weather app predicts tomorrow's high temperature.                                                          |                       |                             |         |
| A model predicts whether a text message is spam or not spam.                                                 |                       |                             |         |